RainAlgae: A Rainwater-Powered, Self-Monitoring Algae Bioreactor

For CO_2 Capture

Team Name: Genesis

Domain: Sustainability, IoT and AI

(hardware *X* software)

One-line summary: A low-cost Chlorella bioreactor that grows in filtered rainwater and shows measured, live CO_2 removal, with automatic protection for the algae .

The problem:

Industrial emissions release large amounts of carbon dioxide, and most capture technologies are expensive and hard to place near small factories. Low-cost algae demonstrations exist, but most only show the algae growing. They rarely give measured, repeatable proof of how much CO_2 is removed, and they cannot protect algae if conditions turn harmful.

Our solution:

RainAlgae is a compact bioreactor that grows Chlorella in filtered, nutrient-enriched rainwater. A CO_2-rich gas stream is diluted and bubbled through the algae under LED light, and photosynthesis absorbs part of the CO_2. For safe testing, we use a yeast-sugar fermentation bottle as a controlled emission source instead of real factory smoke.

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How it works:

Two NDIR CO_2 sensors measure the gas before and after the algae chamber, while pH, temperature and light sensors track the culture. An ESP32 sends the data to a FastAPI backend and a live dashboard. An identical chamber without algae acts as a control, so the difference shows the algae's real contribution.

Methodology:-

1. Algae culture: Chlorella is grown in filtered, boiled rainwater with added nutrients, under LED light on a day and night cycle.
2. Safe CO_2 source: A yeast-sugar fermentation bottle replaces real factory smoke, which can contain toxic gases. The gas is diluted with air to a sensor-safe level and bubbled through the algae.
3. Measurement: Two NDIR CO_2 sensors read the gas before and after the algae. pH, temperature and light sensors monitor the culture. An identical chamber without algae acts as a control.
4. Automation: An ESP32 sends the readings to a FastAPI backend and a live dashboard.
5. Live removal tracking: The system calculates capture % and CO_2 removed in mg per hour, plus a running total.
6. pH safety alert: Too much dissolved CO_2 makes the water acidic and stresses the algae. The system raises an alert if pH moves out of the safe range and pauses the CO_2 supply at critical levels.

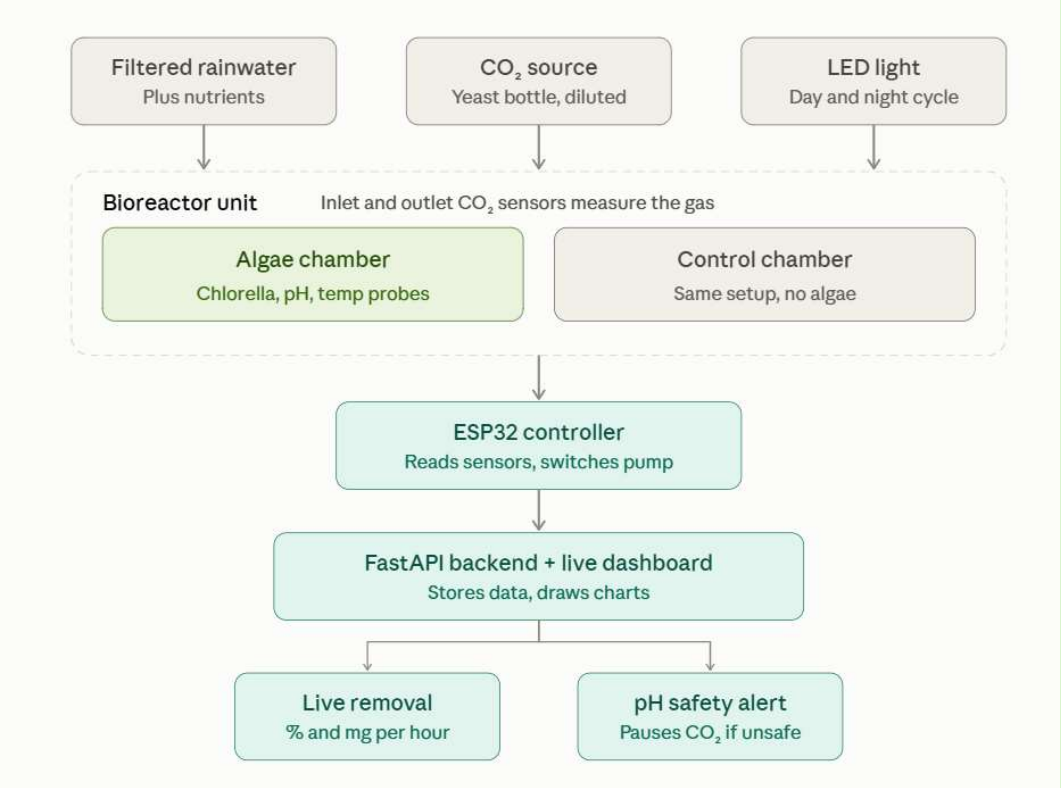
Features:

- *Measured proof: the algae chamber shows consistently higher CO_2 capture than the no-algae control across repeated runs.*
- *Live data: a dashboard showing capture %, mg per hour removed, and a running total.*
- *Protected culture: pH alerts and automatic CO_2 pausing keep the algae healthy.*
- *Reusable design: a low-cost prototype that can be built as modular panels.*

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Proposed diagram of the model:



What makes it unique:

1. **Live removal tracking:** The dashboard calculates capture percentage and CO_2 removed in mg per hour in real time, with a running total and a comparison against the control chamber.
2. **pH-based protective alert:** Excess dissolved CO_2 makes the water acidic and stresses the algae, while heavy photosynthesis can push pH too high. The system raises an alert on either side and automatically pauses the CO_2 supply at critical levels.

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Impact and scalability:

Rainwater reuse lowers water demand, the design is low-cost, and the chamber can be built as modular panels placed near emission points. Harvested algae can be used as biofertiliser, so the captured carbon is put to use and not left to decay.

Important to know about RainAlgae:

- Safe and low-cost: uses a yeast CO_2 source and simple parts, with no toxic gases.
- Honest and verifiable: the control chamber and measured data back every claim.
- Resource-smart: reuses rainwater and can turn harvested algae into biofertiliser, so the carbon is put to use.
- Scalable: modular panels could sit near small emission points.

Future enhancements:

A machine learning model trained on the logged data to predict CO_2 absorption from light and temperature, and a particulate sensor stage for dust and smoke.

Thank You